

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

```

```

df = pd.read_csv("/Titanic-Dataset.xls")

# Display first rows
print(df.head())

```

```

 PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

```

```

      Name      Sex  Age  SibSp  \
0  Braund, Mr. Owen Harris    male  22.0    1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0    1
2    Heikkinen, Miss. Laina  female  26.0    0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0    1
4    Allen, Mr. William Henry    male  35.0    0

```

```

   Parch    Ticket   Fare Cabin Embarked
0      0  A/5 21171   7.2500   NaN        S
1      0    PC 17599  71.2833   C85        C
2      0 STON/O2. 3101282   7.9250   NaN        S
3      0    113803  53.1000  C123        S
4      0    373450   8.0500   NaN        S

```

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```
: df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

/tmp/ipykernel_309/3744086084.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

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```
: df.drop(columns=['Cabin'], inplace=True)
```

```
: df['Sex'] = df['Sex'].map({'male':0, 'female':1})  
  
df = pd.get_dummies(df, columns=['Embarked'], drop_first=True)
```

```
: sns.set(style="whitegrid")
```



```
print(df.info())

print(df.describe())

print(df.isnull().sum())
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 891 entries, 0 to 890
```

```
Data columns (total 12 columns):
```

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	object
4	Sex	891 non-null	object
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	object
9	Fare	891 non-null	float64
10	Cabin	204 non-null	object
11	Embarked	889 non-null	object

```
dtypes: float64(2), int64(5), object(5)
```

```
memory usage: 83.7+ KB
```

```
None
```

	PassengerId	Survived	Pclass	Age	SibSp	\
count	891.000000	891.000000	891.000000	714.000000	891.000000	
mean	446.000000	0.383838	2.308642	29.699118	0.523008	
std	257.353842	0.486592	0.836071	14.526497	1.102743	
min	1.000000	0.000000	1.000000	0.420000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	
50%	446.000000	0.000000	3.000000	28.000000	0.000000	
75%	668.500000	1.000000	3.000000	38.000000	1.000000	
max	891.000000	1.000000	3.000000	80.000000	8.000000	



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```
count    Parch    Fare
mean     0.381594  32.204208
std       0.806057  49.693429
min       0.000000   0.000000
25%       0.000000   7.910400
50%       0.000000  14.454200
75%       0.000000  31.000000
max       6.000000 512.329200
PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            177
SibSp           0
Parch           0
Ticket          0
Fare            0
Cabin          687
Embarked        2
dtype: int64
```

```
df['Age'].fillna(df['Age'].median(), inplace=True)
```

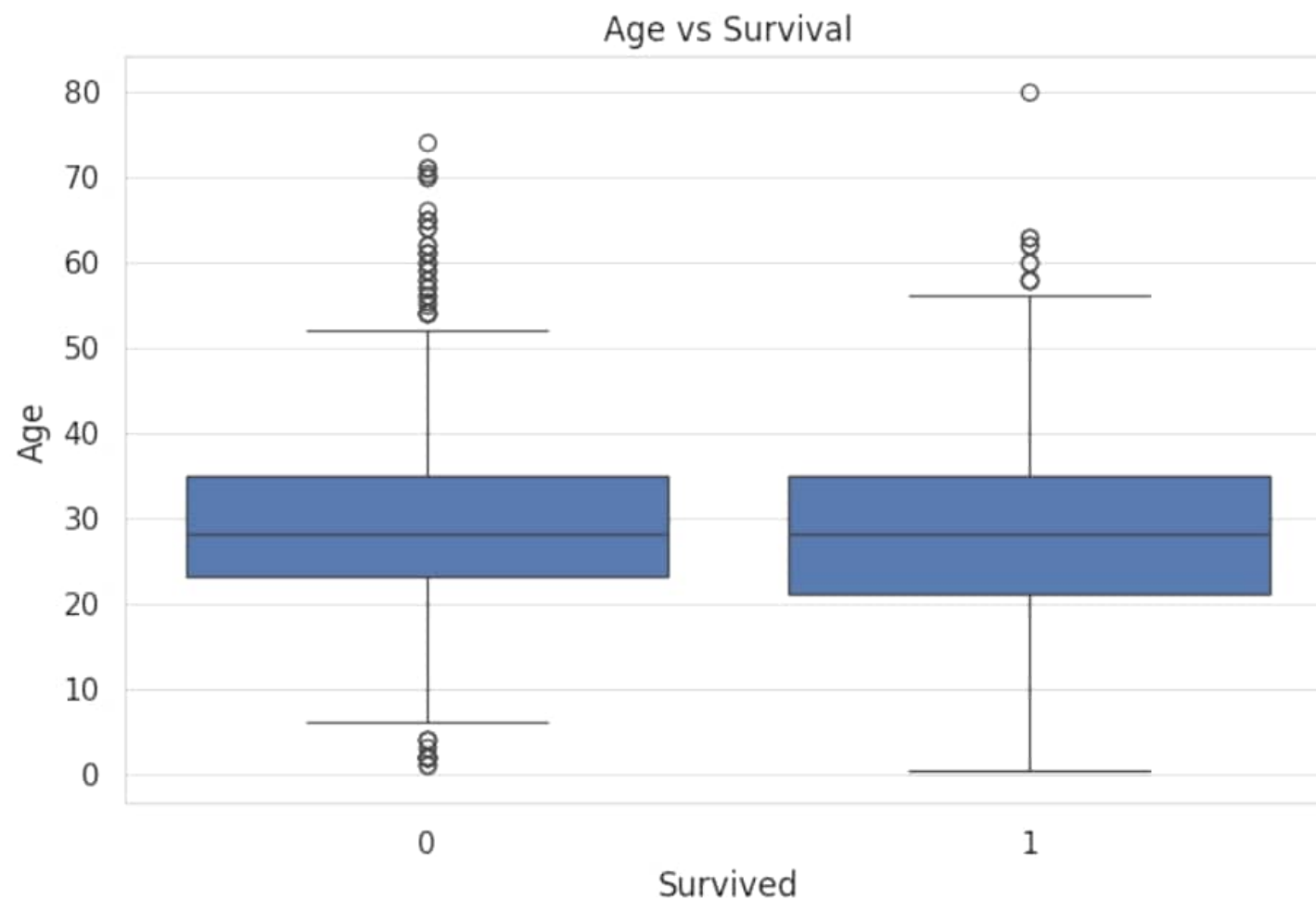
/tmp/ipykernel_309/1933487976.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df['Age'].fillna(df['Age'].median(), inplace=True)
```

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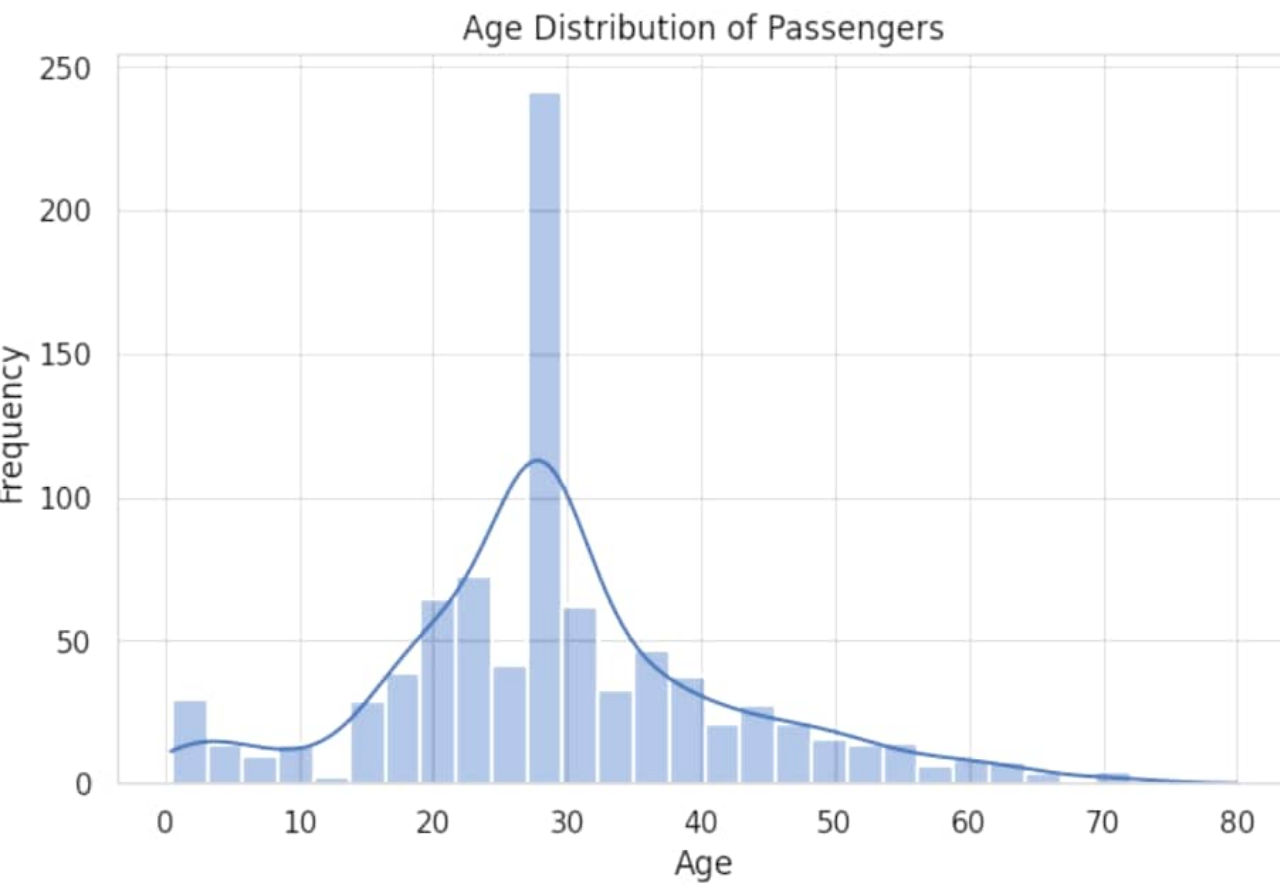
```
plt.figure(figsize=(8,5))
sns.boxplot(x='Survived', y='Age', data=df)
plt.title("Age vs Survival")
plt.xlabel("Survived")
plt.ylabel("Age")
plt.show()
```



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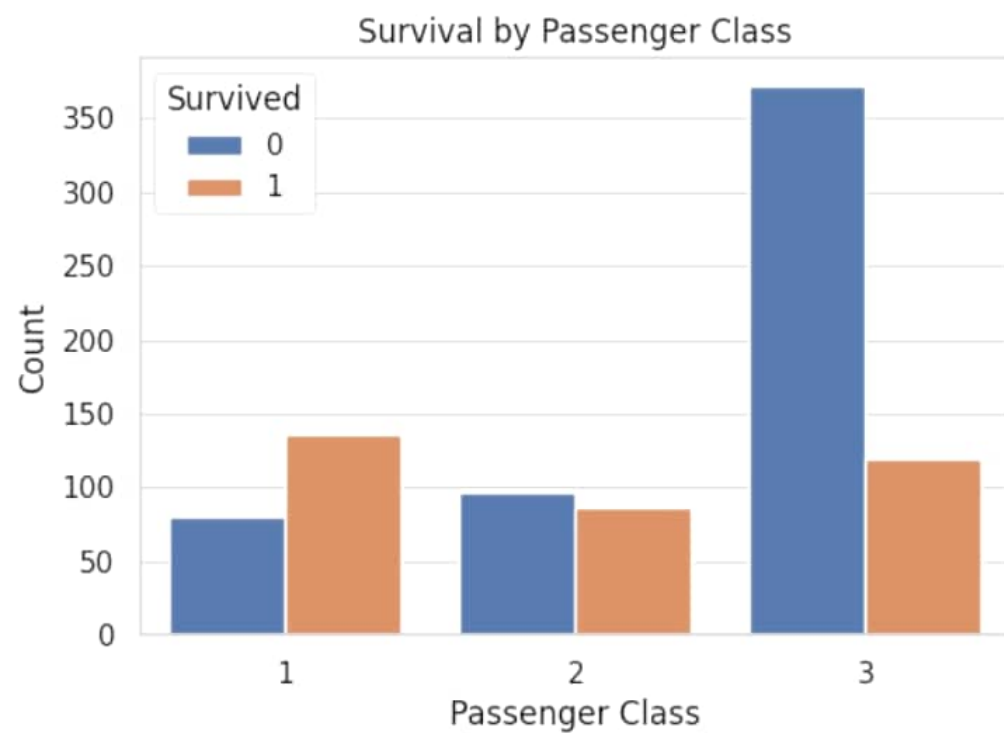
```
plt.figure(figsize=(8,5))
sns.histplot(df['Age'], bins=30, kde=True)
plt.title("Age Distribution of Passengers")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```



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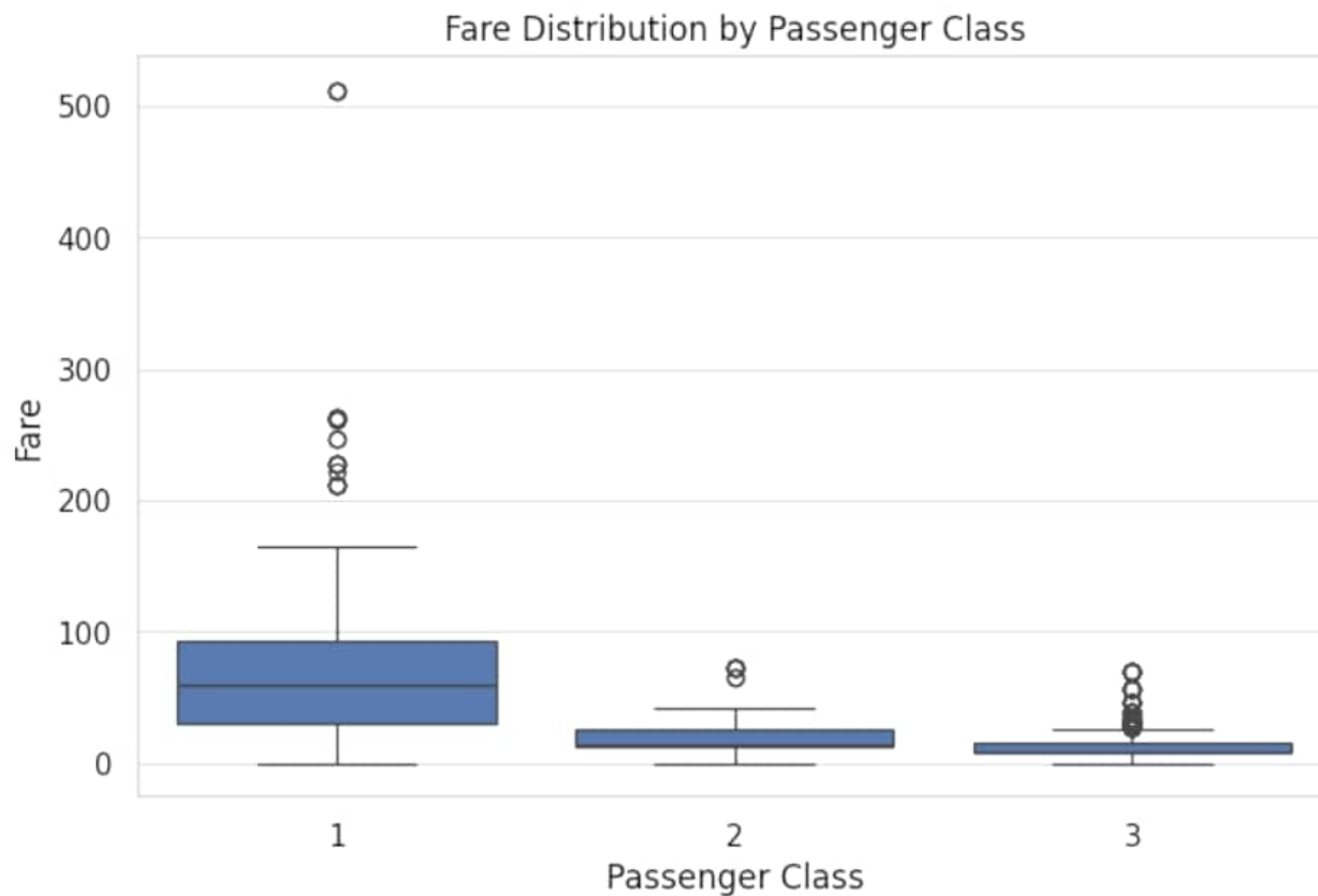
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```
plt.figure(figsize=(6,4))
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title("Survival by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Count")
plt.show()
```



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```
plt.figure(figsize=(8,5))
sns.boxplot(x='Pclass', y='Fare', data=df)
plt.title("Fare Distribution by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Fare")
plt.show()
```



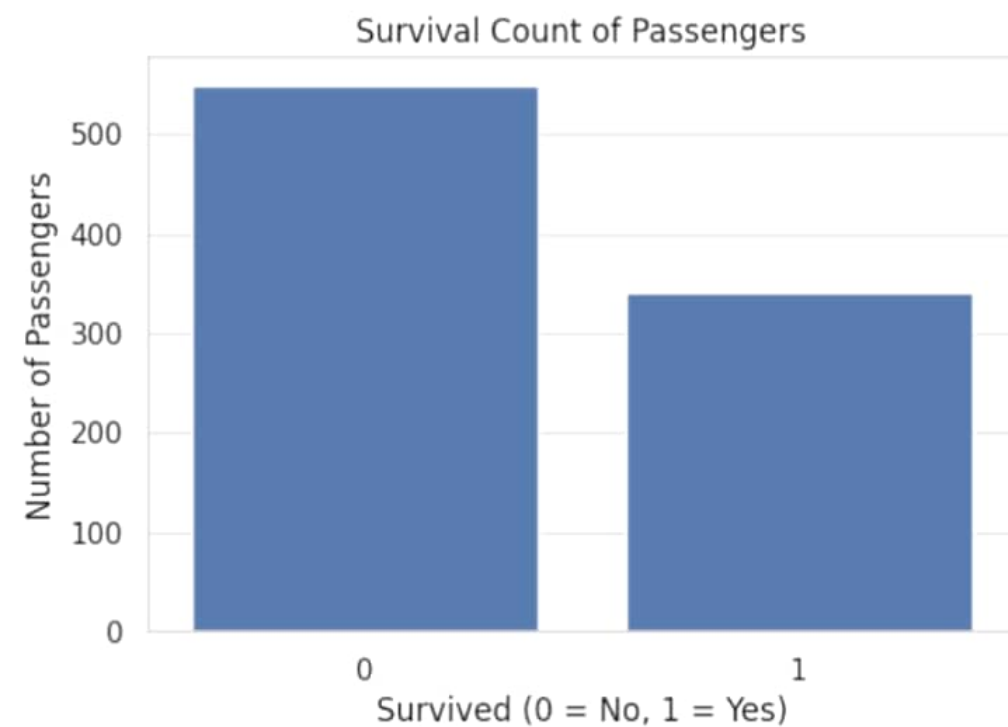
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```
: plt.figure(figsize=(6,4))
  sns.countplot(x='Survived', data=df)
  plt.title("Survival Count of Passengers")
  plt.xlabel("Survived (0 = No, 1 = Yes)")
  plt.ylabel("Number of Passengers")
  plt.show()
```



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```
1]: import matplotlib.pyplot as plt
import seaborn as sns

corr_matrix = df.corr(numeric_only=True)

plt.figure(figsize=(8,5))
sns.heatmap(corr_matrix, annot=True)
plt.show()
```

